CWRUbotix

2026 Technical Documentation

Monitoring and Mitigating the Impacts of Climate Change on Our Water World

Case Western Reserve University
Cleveland, OH, USA

Team Members

Mechanical

Kiran Sah '28 (Mechanical lead)
Amitan Bar-Evan (CEO)
Benjamin Mellick '27 (CFO)
Avery Twomey '29
Caleb Park '28
Olivia Fietkiewicz '27
Nathan Law '27
Simon Fox '27

Software

Shannon Griswold '27 (Software Lead)
Mollie Ackerman '27
Bren Nakata '28
Kelsey VanDenBerg '28
Fotih Turopov ' 28

Hardware

Ilyasse El Asslouj '28 (Electrical Lead)
Joshua Kim '27
Kippei Miki '29
Ryan Shorter '27
Jonathan Callejas '27
Owenkhosi Sibandze '29
Yuan Dong '29
Ivan Hernandez '28
Rebecca Dankulich '27

Mentor

Richard Bachmann

Abstract

In response to MATE's Request For Proposals (RFP), the CWRUbotix ROV team presents Dory and Nemo, a marine monitoring and repair system comprising Dory, the remotely operated underwater vehicle, and Nemo, the vertical profiling float. Dory and Nemo were developed over the course of a year by 22 dedicated employees at CWRUbotix. Together, Dory and Nemo are exceedingly well equipped for polar oceanic operations including coral studies, reef mapping, and traversal through strong polar currents and waves.

The CWRUbotix 2026 design philosophy emphasizes versatility and reliability, with a strong priority on member education through experience. To ensure our products follow the CWRUbotix tradition of excellence in design, all mechanical components of Dory and Nemo are designed, manufactured, and depth tested in-house by CWRUbotix employees. Much of Dory and Nemo's internal electronics are composed of custom PCBs both designed and assembled by CWRUbotix electrical engineers. Each component of the Dory and Nemo integrates with a central surface computer for easy piloting and data access. Dory's design places high importance on reliability, employing techniques proven by previous CWRUbotix ROVs and applying them to new designs, such as Dory's minimized electronics bay and new stereo camera enclosure style.

The CWRUbotix Company

Table of Contents

Project Management

Company Overview

CWRUbotix is an undergraduate student organization of Case Western Reserve University in Cleveland, Ohio, managed and run entirely by students with a mission to explore robotics. The CWRUbotix MATE ROV Competition team is made up of 22 undergraduate students from across all walks of engineering studies and divided into three subteams: mechanical, electrical, and software, with many members contributing to multiple subteams. Each subteam is managed by a subteam lead, an experienced and driven member who is responsible for task delegation and review, ensuring that all tasks are completed thoroughly and in a timely manner. Subteam leads are also in charge of new member onboarding to ensure that new members have the resources they need to contribute to and learn from the team. Subteam leads report to the CEO, who is responsible for full team organization and cohesion, keeping the team's schedule and budget, and coordinating full system integration. The CEO and subteam leads are elected annually by the members of the team. The CEO of the MATE ROV team reports to the CWRUbotix Executive Board, who oversees all competition teams within CWRUbotix and interfaces with university administration. The Executive Board is elected annually by the CWRUbotix General Body, and coordinates organization-wide events, raises and manages funds, and ensures the functioning of each of the competition teams. The CWRUbotix CFO and Safety Manager are members of the CWRUbotix executive board. The CFO is responsible for fundraising, accounting, and purchases across the whole of CWRUbotix, while the Safety Manager ensures that all members comply with our rigorous safety policies and oversees bay equipment and cleanup. This leadership structure allows CWRUbotix to delve into multiple competitive fields by sharing the multitude of administrative responsibilities among the Executive board and team leads.

Schedule

For the last few years, CWRUbotix has utilized an Agile-inspired system consisting of simple milestones with clear deliverables for project scheduling and task management. Where traditional project management systems such as Gantt charts have restricted the team's situational adaptiveness, our system allows for high flexibility in exploring new ideas and handling unforeseen setbacks while ensuring timely project progress. Our timeline consists of four distinct phases: research, design, manufacturing, and testing. The research phase signals the beginning of the project, and is a time for the team to explore new concepts and experiment with significant and ambitious design changes, while simultaneously laying out the high-level design of the project. This phase culminates in a series of subsystem-level design reviews with the entire team. The design phase begins with MATE's RFP, and encompasses the detailed and finalized design of the project along with rapid prototyping and iteration at the sub-system level. The design phase culminates in a detailed design review, at which the team presents their designs to faculty and alumni in order to receive final design feedback prior to manufacturing. The designs are then brought to life during the manufacturing phase, which culminates in a full systems test of the robot, thus kicking off the testing phase. During the testing phase the ROV is iteratively tested at regular pool tests. This is a time for the team to work out bugs, refine project systems, and practice operating the ROV, result-

Project Phase	Task Examples	Outcome	Target Date
Research	Onboarding, design experimentation	Sub-system level design reviews, high level structure.	2025-11-22
Detailed Design	Task analysis, prototyping, integrated CAD	Detailed Design Review	2026-02-28
Manufacturing	Frame assembly, software completion	Full Systems Test	2026-04-26
Testing	Integrated testing, iteration, demo practice	MATE ROV Championship	2026-06-23

Table 1: Project phases for *Dory* and *Nemo*

ing in a highly reliable product and prepared team ready for demonstration at the MATE ROV World Championship.

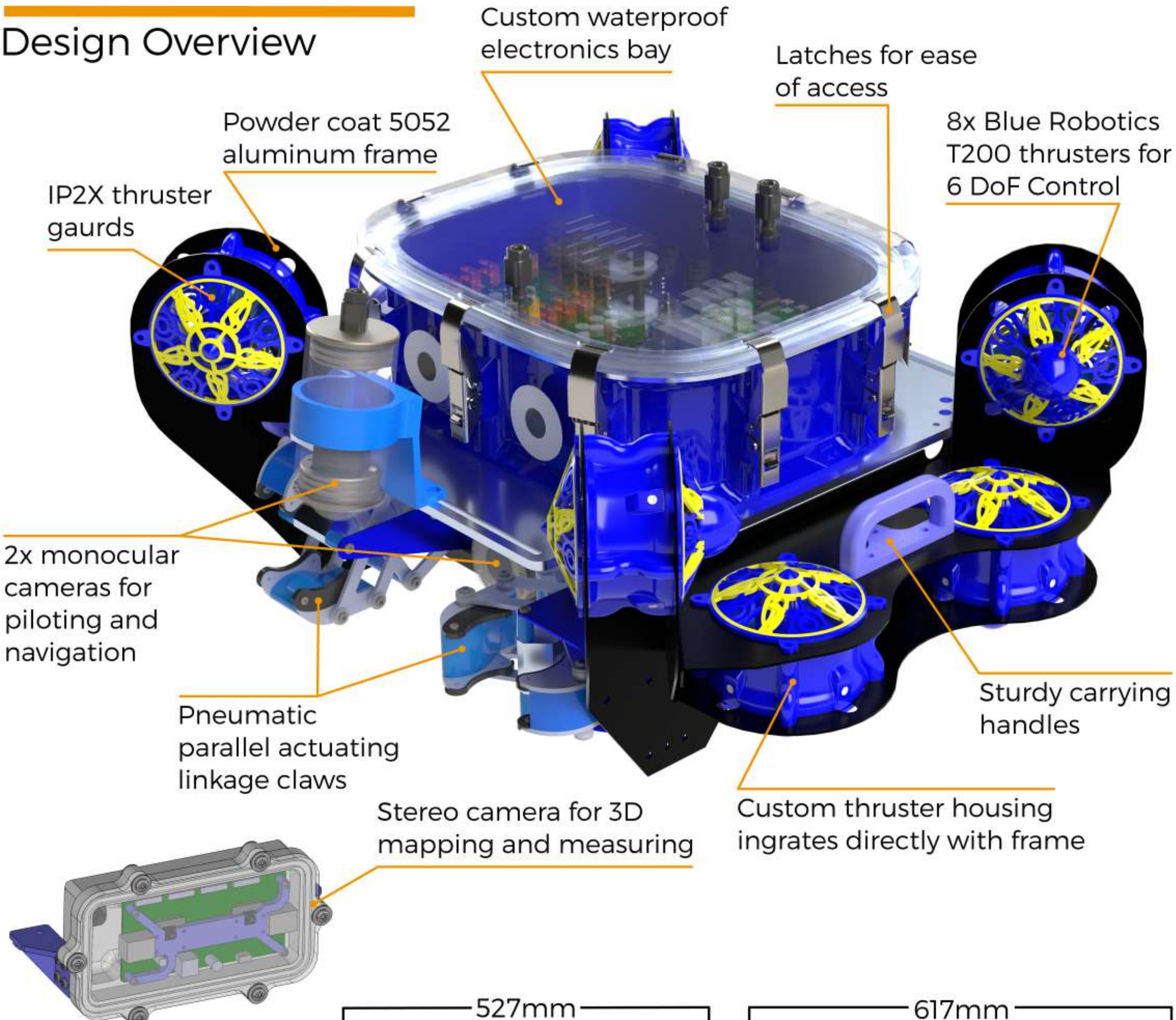
Resources, Procedures, and Protocols

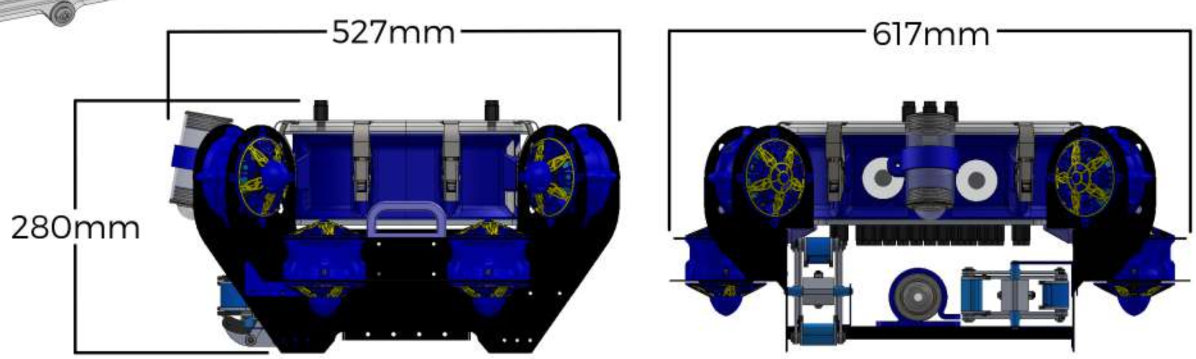
CWRUbotix follows a weekly development cycle, with a full team design meeting every Saturday followed by subteam meetings and a pool test throughout the week. Design meetings are a time for the full team to come together and share progress, discuss design propositions, and plan for the upcoming week. Design meetings are held in a classroom with a projector to allow easy communication and sharing of assets. Meetings are led by the CEO and subteam leads, however all members are encouraged to attend and present their work and ideas. These meetings build consistent communication, encouraging collaboration and integration between subteams, and ensuring that all members have a deep understanding of how their contributions fit into the overall system. After the discussion portion of the meeting, the team has working time with the robot wherein all subteams are present, which facilitates effective system integration and debugging. Throughout the week, each subteam has a regularly scheduled meeting with dedicated time to work on the robot. Additionally, CWRUbotix holds a weekly pool test at the university pool. This frequent testing cycle has proven invaluable for quickly identifying problems and evaluating new solutions.

CWRUbotix uses Discord for all online communication. The CWRUbotix Discord server also serves as a repository of information, with quick access to important links and specifications. CWRUbotix employs Google Suite (including Drive, Docs, and Sheets) for documentation and real-time collaboration, in addition to version control software (GitHub, SolidWorks PDM, and Altium Workspaces for software, mechanical CAD, and PCB CAD, respectively) to ensure all team members have access to the most recent version of the design and can work in parallel on different files. To ensure code reliability, the software team has implemented a code review process and continuous integration which automatically checks all code for syntax and style errors. This year, CWRUbotix implemented a physical kan-ban style board using a whiteboard and sticky-notes to manage mechanical and electrical subteam task progress. Using a physical board allowed members to see at a glance what tasks needed doing and who was responsible for them. The software subteam uses a similar system built into GitHub, which allows for direct integration of their task management and code version control.

Dory

Design Overview





Design Rationale

Engineering Design Rationale

Dory was designed to efficiently complete the tasks in MATE's request for proposals and with reliability and improvements over CWRUbotix's previous ROV iteration, *Fried Rice*, in mind. *Dory's* design changes were made to reflect the team's primary goals of efficiently completing all required tasks, while maintaining high serviceability, ease of usability, and reducing material costs. The team used trade studies alongside data from CAD models, simulations, and previously learned lessons to make informed design decisions for every subsystem. Design brainstorming and feedback was facilitated in full team design meetings. After the feedback of initial iterations was considered and incorporated, prototypes of designs were manufactured and tested in-house to provide additional feedback for further reiteration to improve the quality and efficiency of the components.

Systems Approach

The mechanical, electrical, and software systems of *Dory* were developed in parallel, with frequent intercommunication facilitated by full-team design meetings. Systems-wide decisions, such as the design alterations required for *Dory's* electronics bay size reduction, were discussed and agreed upon at these full-team meetings, ensuring no design inconsistencies were overlooked. Additionally, the electrical team provided 3D models of their PCBs, which were incorporated into the CAD models of mechanical systems to ensure that reasonable tolerances of system components could be achieved and that all the components of the ROV were compatible. This model allowed potential intersections and conflicts between parts to be resolved before manufacturing any physical prototypes, saving material and development time. The model made the integration of the ROV systems more efficient and reliable, as the requirements of electronic and mechanical system interfaces (i.e. penetrator placement, PCB placement within the electronics bay) were able to be identified and design modifications were able to be made to adapt to the requirements of each system without much delay.

Vehicle Structure

Dory's primary structural components are a sheet metal frame and a watertight electronics bay (described below). This design prioritizes low cost and high reliability under environmental influences. To maximize power efficiency and maneuverability, Dory is designed to be neutrally buoyant and hydrostatically stable, meaning that it will naturally remain stationary and upright without requiring input from the pilot. This requires that the mass and volume of the ROV be carefully tracked during the design process and fine-tuned after manufacturing (see Buoyancy and Ballast), The envelope size of Dory occupies a substantial volume to increase buoyancy and for an even placement of the thrusters. Per MATE's RFP, Dory was designed to weigh less than 18 kg in air, allowing for handling by a single person. While a heavier ROV is less maneuverable, Dory's considerable size and weight also makes it less prone to environmental factors, such as currents. However, Dory's mass has been reduced from Fried Rice's design, in large part by a reduction in the volume of the electronics bay resulting in less need for ballast. Due to the design changes in Dory's structure, the overall cost has also been reduced. The costs of a larger size include reduced acceleration and top speed (due to inertia and drag) as well as reduced access to tight spaces. The former downside is mitigated by a powerful propulsion and electrical power system, while the latter is mitigated by ample pilot practice. The team found in testing that Dory is still small enough to easily navigate its environment and complete the required tasks, and that travel speed is of less importance than maneuverability, dexterity, and manipulation capability.

Frame

The frame is constructed from bent sheet metal parts and employs a consolidated design that reduces both the number of fasteners and the weight of the ROV. The frame is composed of 7 individual parts, as opposed to Fried Rice's 15 parts. The frame is additionally designed to allow for tools to be mounted to the ROV efficiently and can be used for quick tool swaps. The material selected for *Dory* is 1.6mm (1/16 in) aluminum 5052 sheet metal, as it is ideal for bending due to its formability while still being of a reasonable cost and a reduced density. The aluminum additionally has a good strength-to-weight ratio, which allows for the weight budget to be allocated to the rest of the ROV. The fasteners used on the frame are 304 stainless steel machine screws and rivnuts, which have reduced the cost from Fried Rice's 316 stainless steel fasteners while maintaining a reliable corrosion resistance and durability. One of the major drawbacks of using aluminum with stainless steel fasteners is the risk of galvanic corrosion, which occurs when dissimilar metals are in contact or close proximity in an electrolytic solution. In order to mitigate this, stainless steel rivet nuts are installed in the frame components, which are then powder coated to protect the interface from water.

The handles of the ROV, as well as the thruster guards and shrouds, are 3D printed with PETG filament for PETG's degradation resistance over PLA.

Figure 1: Exploded view of *Dory*'s frame

Electronics Bay

The electronics bay (e-bay) walls are manufactured from PETG utilizing an Anycubic Kobra 2 Max Fused Deposition Modeling (FDM) printer [1]. This e-bay represents an iterative advancement of previous CWRUbotix designs, specifically refined to optimize internal component layout and integration. The structure employs a rectangular "squircle" geometry—a hybrid between a square and a circle—to maximize internal volume while maintaining the structural integrity required to withstand hydrostatic pressure through rounded exterior surfaces.

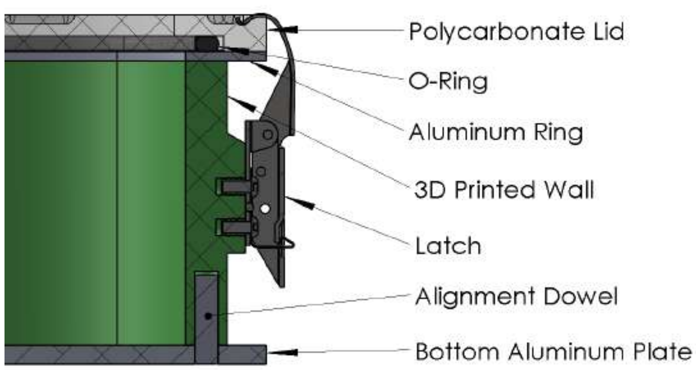


Figure 2: Construction of the e-bay

A notable change from last year's ROV is a reduction in the dimensions of the enclosure, as the length has been shortened from 328 mm to 268 mm, which translates to an 18,000 mm^2 reduction in interior space and an approximate 600 gram reduction in weight, with additional weight reduction resulting from less needed ballast. A priority for this iteration of the ROV was additional space on the base plate to install camera enclosures and special manipulators, which proved to be an issue with the size of the camera enclosures used for the 2025 competition. To improve the accessibility for installation of these components and for long-term flexibility towards accommodating different special manipulators, the decision was made as a team to reduce the footprint of the e-bay.

By 3D printing the enclosure with PETG, the team achieved a custom geometry at a low material cost. The e-bay is mounted atop a 6 mm (1⁄4 in) aluminum base plate, which serves as the primary heat sink for the internal power and control systems. The external dimensions of the final design measures 328 × 268 mm, providing a 300 × 240 mm internal rectangular squircle footprint. With an interior height of 100 mm, the enclosure provides sufficient volume for substantial components even with the size reductions, including the 48V power conversion system, flight computer, electronic speed controllers, and Ethernet switch.

The e-bay is integrated onto a waterjet-cut 6061 aluminum base plate. Aluminum was selected for its high thermal conductivity and excellent strength-to-weight ratio, fulfilling a dual role as a structural frame element and a heat sink for the power converter. This base plate includes integrated mounting tabs for frame attachment, along with dedicated slots for the flexible installation of cameras and mission-specific manipulators.

To validate structural integrity and optimize wall thickness, finite element analysis (FEA) was conducted using the SolidWorks Simulation package (figure 3). This enabled the team to simulate the effects of water pressure across all external surfaces. Simulations were performed at a depth of 7 m, equivalent to approximately 70 kPa of pressure, as illustrated in Figure 4. The analysis identified a peak stress of 8.8 MPa. Given the 47 MPa yield strength of PETG, the design maintains a factor of safety of 5.3.

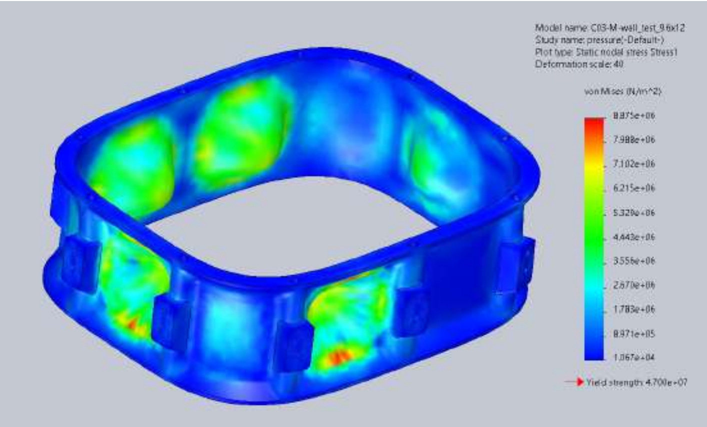


Figure 3: FEA simulation of the e-bay wall

The upper enclosure lid is machined from clear polycarbonate using a Laguna CNC router. This transparency allows operators to perform visual diagnostics of internal systems without compromising the seal. The lid features a large O-ring for waterproofing and is secured by 10 latches, facilitating rapid, tool-free access to the electronics. To validate that the latches fully seal the lid, FEA was performed through the Solidworks Simulation package (figure 4). Simulations were performed for the location of the 10 latches on the enclosure lid each exerting a force of 480 N to reflect the weight capacity of the latches. The peak displacement of the lid along its edges was 0.011 mm, which was more than sufficient in indicating that this layout of latches would fully seal the e-bay.

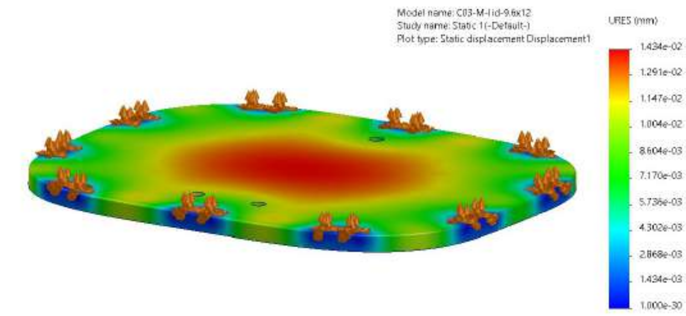


Figure 4: FEA simulation of the e-bay lid under force of the latches

For cabling, the e-bay incorporates eighteen Blue Robotics penetrators through the bottom plate to service thrusters, sensors, and actuators. Additionally, three penetrators are located in the lid to accommodate the power and Ethernet lines for the tether.

Control and Electrical Systems

The ROV's electrical and control architecture is composed of two primary elements: the surface based control station and the onboard electronics housed within the ROV. These elements are connected via a tether. The onboard system is split into two subsystems: the power system, which handles the conversion and distribution of electrical power received from the surface, and the control system, which manages communication with the surface station and governs operation of all onboard components. See figure 8 for an SID of the electrical system.

Power Electronics

The ROV is powered by a power supply box on the surface, which sends power to the ROV at 48V. However, Dory's thrusters run at 12V, so the power from the tether must be stepped down before use. To solve this challenge, CWRUbotix developed a custom in-house 1200 watt power conversion solution: ScuPSU-Mini. This year, the team redesigned the power electronics by splitting last year's single-board solution into two boards: The ScuPSU-Micros. This split was motivated by two factors: first, the team reduced the electronics bay size and explored layouts with stacked or side-mounted boards to maximize space efficiency, but ultimately determined that the space gained would be offset by the need for additional cooling to replace the heat sinking provided by the e-bay bottom; second, with the team rebuilding its knowledge base, this redesign served as an accessible initial project while maintaining the same electrical architecture. Using two onboard buck converters, ScuPSU-Micro efficiently converts the 48 V DC supply from the tether to 12V DC to power the ROV's thrusters. In addition, the board includes integrated voltage regulators to supply stable 5V and auxiliary 12V outputs for control electronics and other components. ScuPSU-Micro is significantly smaller and less expensive than the CWRUbotix's previous solution, which was purchased from a local electronics company. In addition to converting the voltage, ScuPSU-Micro distributes power to the ROVs eight electronics speed controllers (ESCs). Each ESC connects to a high-current XT60 connector and a DuPont header for PWM signals. Though PWM signals originate from the flight computer and not ScuPSU-Micro, routing these signals through the power board greatly simplifies ESC wiring, allowing each ESC to connect directly to a single board. All eight PWM signals are carried from the flight computer to ScuPSU-Micro via a single ribbon cable. To reduce expenses associated with international manufacturing tariffs, the board was entirely assembled in-house.

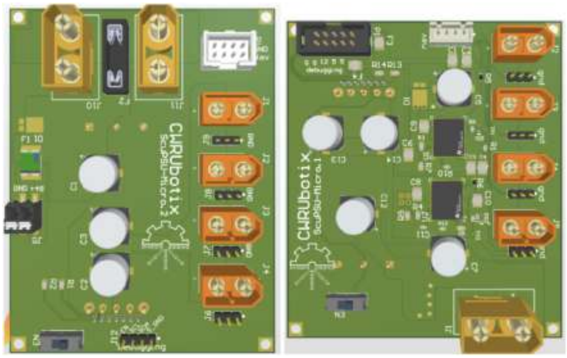


Figure 5: ScuPSU Micro

Figure 6: Custom Navigator

Navigator

The Navigator serves as the flight computer for Dory, handling all real time control and motion related tasks critical to stable underwater operations. It features onboard sensors including a magnetometer and compass for navigation and streamlined integration for a leak detection system. The Navigator outputs PWM signals to control the ROV's thrusters, translating high level movement commands into precise motor instructions. PWM is additionally used to control two relays wired to solenoid valves outside the vehicle in order to control manipulator actuation. By processing sensor data and managing the ROV's response to its environment, the Navigator ensures stable, responsive maneuvering across all six degrees of freedom. This board plays a crucial role in determining how the robot moves, responding to pilot input, and maintaining its position, enabling both manual and autonomous operation with high reliability. A long term project for the team involves designing a custom simplified Navigator board (figure 6) to interface directly via a SODIMM slot with a motherboard, which would also contain a Raspberry Pi Compute Module. While currently Dory is using a COTS Navigator board mounted as a hat to a Raspberry Pi 4, eventually all these components will transition to this custom integrated system.

Control Box

To improve on operational efficiency, the ROV surface control box replaces previous CWRUbotix ROVs' cable sprawl with a single, easily transportable unit. The control box accepts AC power through an IEC C14 socket, which is directed by a consumer power strip to the surface laptop power supply and (using a built-in converter) over USB to the surface router. We chose to purchase a power strip for our high voltage wiring to ensure robustness in this critical system. The laptop power supply powers a central laptop dock which provides port expansion over a single thunderbolt cable for the router, a built-in monitor, and the profiling float surface transceiver. The control box also includes pass-throughs for 48 VDC and pneumatics, with an inline emergency stop button for DC power and a pressure regulator and readout for the pneumatics.

Figure 7: The control box

Tether/Tether Management

Dory's tether is engineered for reliability, emphasizing flexibility and durability. It incorporates two power cables, a CAT6A Ethernet cable, and a 6mm (1/4 in) polyurethane pneumatic line. The power cables are 10 AWG UL Standard 1426 marine grade wires engineered to be lightweight and flexible, with the positive wire linked to a 25A fuse positioned between the ROV and control box. The Ethernet cable is outdoor rated for enhanced durability. Strain relief anchor points are positioned on both sides of the tether, securing to the frame of the ROV and the control box. Additionally, flexible strain relief cones (3D printed in TPU) are fitted around the penetrators in the lid of the e-bay to safeguard the cables from bending. To maintain the tether clear of the ROV's path, several blocks of buoyancy foam are affixed to the tether. The tether is engineered to be approximately 15m long. This is sufficient to reach all of the tasks in the pool yet short enough that there is only a minimal voltage drop of approximately 1.9V across the power wires when the maximum current of 25A is drawn.

During operation of the ROV, a tether manager is assigned to follow CWRUbotix's tether management protocol. While transporting the ROV outside of the pool, the tether manager carries the tether and ensures that it does not become a tripping hazard. Before deployment, the tether manager coils the tether on the pool deck. During operation, the tether manager pays out and reels in the tether as the ROV moves, maintaining that there is no tension in the tether while ensuring that there is no excess slack in which the ROV could become entangled.

SID

ROV full load Amps (FLA) in water = 21A

Fuse size selected based up FLA = 25A

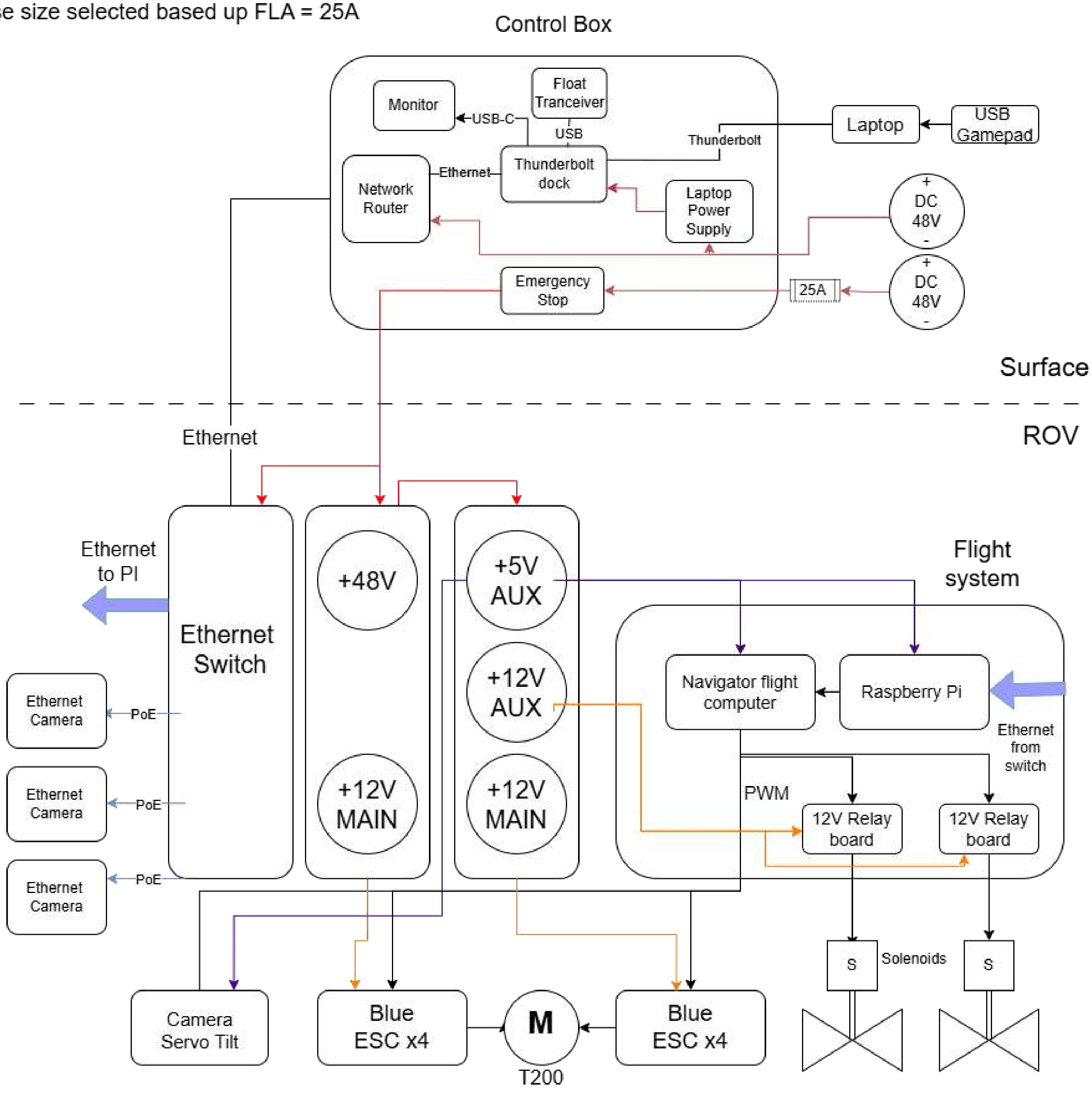


Figure 8: Electrical system SID

Software

Dory's control system is split into three subsystems:

1. the surface system, which is hosted by a laptop on the surface,
2. the ROV system, hosted on the Pi CM5 module onboard the ROV,
3. and the float system, a custom RP2040 float board and an Adafruit Feather transceiver board.

The surface is connected to the ROV over an Ethernet network managed by the surface-side router. Using Ethernet for communication over both tethers minimizes camera feed latency (versus USB) and tether size (versus many analog connections).

Software SID

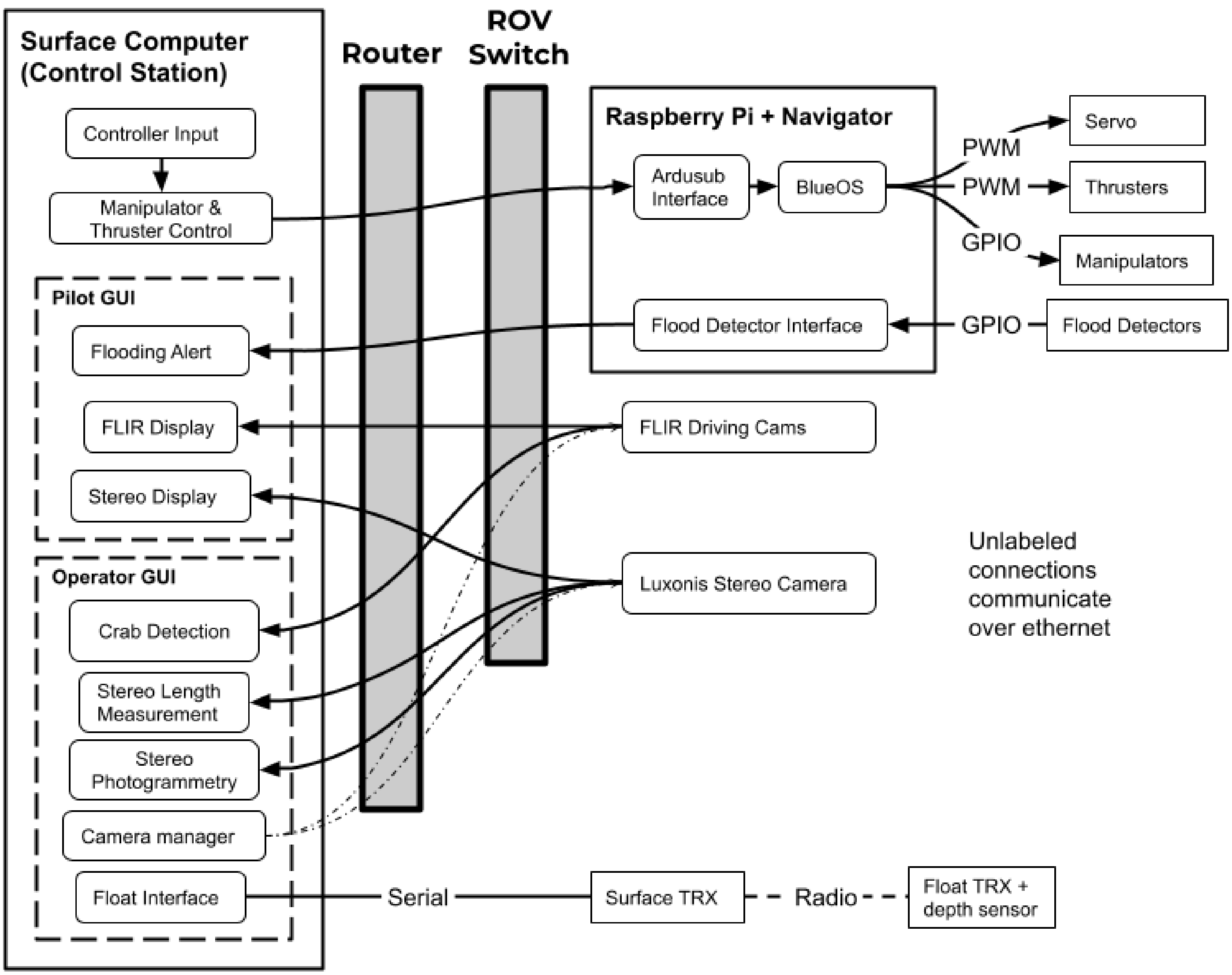


Figure 9: Software SID

Communication

For communication over the tether, ROS 2 is used for all latency-insensitive messages, such as graphical user interface (GUI) actions, as (1) its highly structured design encourages well-organized networks with strongly typed messages, and (2) our previous iterations of the Dory system were designed to integrate with ROS. The ROS paradigm supports easy codebase organization by splitting what would otherwise be monolithic single file control logic into discrete, parallelizable units called "nodes." By organizing our codebase around ROS nodes, CWRUbotix was able to create a GUI with the highly maintainable model-view-controller paradigm, completely decoupling robot state management from its graphical representation.

Pure Mavlink messaging was chosen for pilot thruster instructions to minimize latency. A communication concern we had this year was reliability when communicating with our manipulators. Previously, we communicated with the manipulators by sending a ROS message over the tether, and then an i2c message would be sent to control the manipulators. Due to some reliability issues we experienced with the i2c message, we decided to change our manipulators to be controlled via pure Mavlink, like the thrusters.

AI Crab Detection

To be able to identify the crabs in the sample, we decided to use an Ultralytics YOLO object detection model. We trained this model using manually labeled videos that we took from Dory while underwater and in various lighting conditions so that the model will be able to better identify the crabs in different conditions. This task is completed through a tab on our operator GUI. This tab shows the video from our downward facing camera and can be paused when the crabs are in view to run the detection model on that specific frame.

Photogrammetry and Measurement

Both Photogrammetry and Measurement capabilities were implemented using Depth AI, Open3d, and footage captured from two Luxonis OAK-FFC OV9782 W sensors and a OAK-FFC 4P PoE stereo camera main board [4]. Depth AI is used to create a point cloud of an image taken from the stereo camera stream. Open3D is then used to create a visualization of these point clouds that the operator can use to perform the photogrammetry and measurement tasks. Depth AI was used because it was developed for these cameras by Luxonis and the cameras were already using Depth AI to display the footage for the different cameras on the GUI. Open 3D was used because it can easily work with a Depth AI point cloud and allow us to visualize the point cloud with it.

To perform photogrammetry, we capture multiple point clouds and use RANSAC to combine them. We then create a mesh of this combined point cloud to construct a 3D model. To take measurements, we use Open3d to receive points of interest on the point cloud. The operator uses the Open3D point cloud visualization to select two points. Those points are returned by Open3D and then the Euclidean distance between them is calculated. Both these capabilities are added to the operator GUI so that the operator can easily perform these functions.

Propulsion

While the priority in the design of the thruster placement was in balancing the performance and the mission requirements with the cost of the thrusters. Dory is propelled by eight T200 [3] thrusters from Blue Robotics. The reliability of the thrusters have been tested and documented thoroughly by the manufacturer and have additionally been proven by previous vehicles from CWRUbotix and many other companies. Moreover, the T200s are the most affordable option among reputable vendors who provide testing information. The eight thrusters are arranged with four vertical and four horizontal thrusters. The arrangement of thrusters allows the pilot to operate in all three translational axes (surge, sway, and heave) and torque about the three rotational axes (yaw, pitch, and roll). All six degrees of freedom can be achieved with fewer thrusters, although eight thrusters allow for greater efficiency, as the thrusters responsible for horizontal movement are separated across the length and width of the vehicle. With the current thruster configuration, all the thrusters are also able to be mounted to the side of the vehicle and keep the front and the back of the ROV clear for cameras, manipulators, and tools, and the control system and power distribution across the thrusters is also simpler.

Buoyancy and Ballast

Dory is designed to be neutrally buoyant and hydrostatically stable when upright. Neutral buoyancy requires that the mass and volume of the ROV be precisely balanced so that its weight is equal to the buoyant force when submerged. Hydrostatic stability is achieved when the center of mass of the robot is directly below the center of volume, at which point the net weight and buoyant forces are collinear, and any change in orientation creates a restoring moment which returns the vehicle to its equilibrium position. In order to achieve these goals, all components of the ROV were modeled in CAD with accurate mass and density. The total mass and enclosed volume, as well as center of mass and center of volume, were determined from the model. From this information, the company determined an estimate of the amount of ballast required and where it should be placed to achieve the desired condition. Using this estimate as a starting point, engineers trimmed the buoyancy of the final ROV by placing it in water and adding, removing, and relocating ballast until the ROV sits level and does not tend to sink or float.

Payload and Tools

Linkage Claws

Dory has two manipulators that are designed to manipulate objects that do not require to be rotated. Each manipulator is mounted to the sheetmetal frame just underneath the e-bay, on the left and right sides of Dory. The right manipulator is configured to grab vertical items, and the left manipulator is configured to grab horizontal items relative to the forward facing camera that sits in between the two manipulators.

Figure 10: Parallel actuating linkage claw

The vertical manipulator shown in figure 10 is a clean sheet design that uses a pneumatically actuated parallel linkage mechanism designed for simplicity and reliability. This new design is motivated by the issue that the non parallel jaws of the previous generation of manipulators would push objects out before the manipulator fully closed. The primary design criteria of the vertical manipulator were parallel jaws, reliability, simplicity, small form factor, and large acquisition area. With these constraints in mind, the manipulator could not be another traditional parallel linkage mechanism because of the gears and/or linear slides that are prone to failure and difficult to manufacture. These shortcomings of traditional parallel linkage mechanisms led to the development of a parallel linkage mechanism that uses only simple rotational pivots and is pneumatically actuated for reliability, instead of underwater motors that require advanced electric and software control, on top of waterproof housings.

Figure 11: Reused horizontal linkage claw

The horizontal manipulator shown in figure 11 is an example of our previous generation of manipulators, due to the unique and unforeseen challenges of the horizontal manipulator. One of the main operations performed by the horizontal manipulator is picking up objects off the floor. The previous generation of manipulators have two sets of two prongs that come together to grab an item. We have found that these prongs are exceptional at scooping items off the floor. Iterations over this new design will continue in order to reduce their size and usability with ground items.

Lego Claw

In order to remove biofouling from the observatory (spin inner plastic sheet three times), a claw attached to a servo in a Blue Trail Engineering waterproof housing is used. The claw has inward filets to allow the knob to be easily guided into the claw to rotate. Since the claw obstructs the horizontal manipulator on the robot, there is a snap on and off mechanism connecting the claw to the waterproof servo, allowing the props manager to easily remove or reattach the claw when needed. This snap on and off mechanism is inspired from the way the Lego Bionicle toy joints allow children to snap on and off joints from the toy.

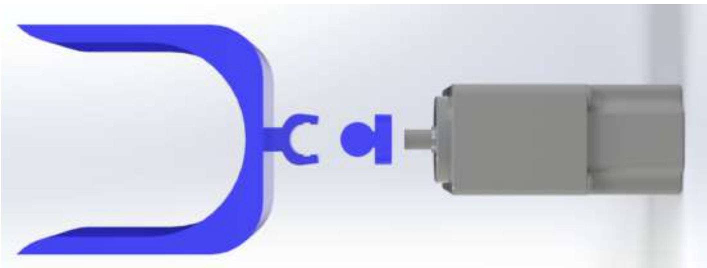


Figure 12: Lego claw

Onboard Cameras

Dory's camera system was designed to meet the needs of both piloting and computer vision-based mission tasks while balancing manufacturability and serviceability. The camera system consists of forward-facing and downwards-facing monocular cameras towards the front of the robot and a stereo camera on the rear of the robot. The two monocular cameras are positioned to ensure that the linkage claws can operate comfortably within each camera's FOV (field of view), aiding the pilot in aligning the manipulators with underwater objects. The forward-facing camera allows for horizontal vision and for visually determining the depth of the ROV, and the downwards-facing camera provides the pilot with a top view of the testing environment for easier navigation. The stereo camera can be used for both piloting and taking measurements of icebergs and coral. With this camera system, Dory is able to perform all mission tasks with only three cameras. All cameras operate over Ethernet, these industrial IP cameras were chosen for Dory because they meet our latency requirements, do not require additional cables in the tether, and do not consume compute resources on the onboard computer.

The monocular cameras are Blackfly S GigE FLIR cameras [2], graciously donated by Teledyne Marine, and are high quality machine vision cameras designed for industrial use. Each FLIR camera is connected to the e-bay by a single Cat 6 Ethernet cable, which carries both data and power. A Power over Ethernet (PoE) capable network switch in the e-bay injects 48 V power from the tether to power the cameras. It also routes network traffic from the cameras to the Cat 6 Ethernet cable which runs up the tether to the surface managed router, where the streams are routed to the surface laptop and displayed to the pilot. This streaming solution results in a glass-to-glass latency of less than 100 ms. A ROS camera management node provides ROS services to toggle individual camera streams, keeping any unnecessary streams off the network to avoid wasting bandwidth.

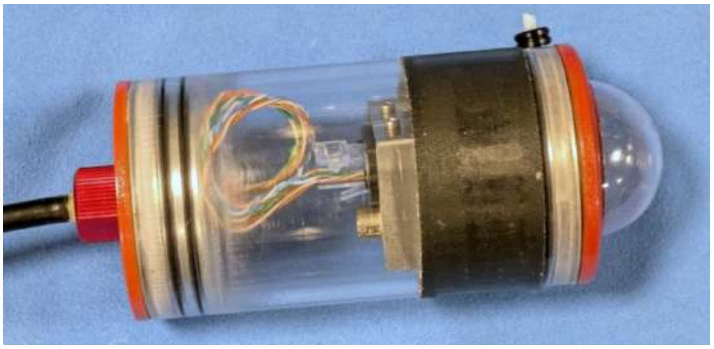


Figure 13: Monocular camera

The stereo camera system is designed to take consistent measurements of the iceberg and coral structures (using stereo vision to generate a disparity map from which 3D position may be derived). Luxonis provided CWRUbotix with a discounted Luxonis OAK-FFC 4P PoE stereo camera main board and two OAK-FFC OV9782 W sensors [4] in 2025. Using separate Luxonis PoE and camera sensor boards allows for configurable baseline distance between stereo camera sensors (as opposed to all-in-one stereo options like the OAK-D Lite), which in turn improves disparity map accuracy for large objects. The Luxonis OAK lineup also provides onboard compute, allowing us to run image processing pipelines directly on the camera board, offloading image rectification and freeing up the Pi's resources for other tasks. Using this system, we are able to perform every mission task with only three cameras.

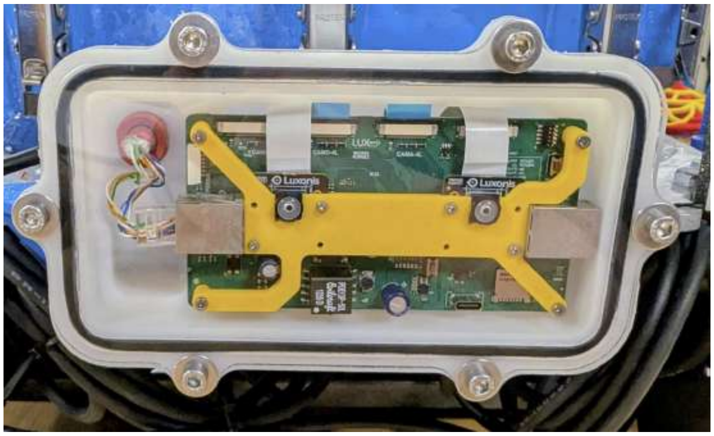


Figure 14: Stereo camera

An air-filled enclosure for each camera was manufactured completely in-house, with specialized enclosure styles for the monocular and stereo cameras. The monocular cameras are housed in a tube-style enclosure (figure 13), with durable custom machined aluminum end caps, a polycarbonate tube and a custom vacuum formed dome which reduces distortion to the lens (as compared to a flat interface). The end caps are secured to the tube with a quick-release locking cord for serviceability. The stereo camera enclosure (figure 14) uses a box CNC routed from an ABS block which interfaces via a face-style seal with a clear polycarbonate lid. Previously, this camera was also housed in a tube-style enclosure, which caused significant distortion which in turn disrupted the camera's ability to create its 3D point cloud. This new enclosure instead uses a flat and clear polycarbonate lid placed directly against the camera lens in order to eliminate this distortion. The lid is secured via six bolts, and can be easily replaced should it become scratched.

Build vs. Buy, New vs. Used

The factors that contribute to the decision to build components in-house or to buy components are the experiential value, reliability, and the cost of the part. The purpose of CWRUbotix is to provide members with opportunities to gain hands-on engineering experience, and so we design and manufacture everything we can in-house. When making the decision to build in-house or outsource, the reliability of the part, time, and monetary costs are all considered. If a part is particularly complex or expensive to make (i.e. cameras, thrusters, and cable penetrators), we will likely buy the component rather than build it. That being said, the experiential value and learning opportunities of designing and building parts in-house will often outweigh the time cost and potential risk, even for critical components such as enclosures, the electronics bay, and manipulators. The decision of whether to reuse components is dependent on whether the part was purchased or built, and the value associated with using a new part. Purchased parts such as cameras and thrusters are often reused to reduce waste and monetary costs, while custom built parts, such as electronics bay and enclosures, are often remade year-to-year for both the experiential value and so the designs can be iterated and improved upon.

Safety

Personnel Safety

The safety of team members is CWRUbotix's highest priority, along with the safety of the fellow design teams in CWRU's makerspace, Sears think[box]. CWRUbotix team members adhere to industry-standard safety practices at all times, and the team has a dedicated Lab and Safety Manager who enforces these safe practices, ensures that the workspace is clean, as well as maintains and makes available safety equipment. As per think[box] policy, the Lab and Safety Manager grants deputy status to trusted long-time members, who receive unlimited bay access and are responsible for supervising the space whenever it is open; new members are not permitted in the bay without at least one deputized person present, and deputies serve as the primary contact with think[box] in the event of an incident. The team fully complies with think[box]'s safety guidelines; for instance, when lead was discovered in the bay, qualified professionals from the building performed the removal after the team collected the material. In the lab, first aid kits and fire extinguishers are placed in convenient locations with good visibility. The team prioritizes injury prevention over treatment, but in the rare instances where injuries occur, first aid is administered promptly using the readily accessible medical kit. All flammable materials are stored in the fire cabinet unless they are being actively used, and aisles are kept clear. Team members in the lab are required to wear safety glasses and closed toed shoes, and wear additional safety equipment when necessary. Inexperienced members always operate potentially hazardous tools, such as soldering irons, under the supervision of more experienced members.

Equipment Safety

There are several safety features implemented on the ROV. The thrusters are surrounded by IP20 guards that prevent objects from contacting the blades of the propellers, protecting operators and wildlife and avoiding objects becoming tangled in the thrusters. All metal components of the robot are deburred to prevent injury when handled. The strain relief system ensures that the ROV can be safely held by the tether without damage to the cables, preventing electrical faults and electrocution. Pressure gauges are used to measure the pressure in the pneumatic tube, and there are mechanical emergency valves in the event that the pressure exceeds 40 psi.

Operational Safety

During testing, the ROV is confirmed by a team member to be powered off and the pneumatic line is depressurized between tests and before touching the electronics. The ROV construction safety checklist and ROV operation safety checklist are detailed in Appendices A and B.

Critical Analysis

Testing & Troubleshooting

Our testing approach is as follows: to frontload as much risk as possible by testing systems and subsystems frequently and under demanding conditions relevant to the competition tasks. This de-risking often involves concurrently testing systems and subsystems at different levels of integration.

We first perform unit testing at the lower subsystem levels for all enclosures and actuators. For all but our largest mechanical enclosures and actuators, this involves hydrostatic testing to a minimum Factor of Safety (FOS) of four. (The maximum depth of the pool during competition is three meters, so the hydrostatic chamber simulates a depth of up to 20 meters). For our main electronics bay, the size requires a different testing methodology in which we attach ballast, and sink it in the deep section of a pool to 3.05 meters (10 feet) for one hour. Inside the electronics bay during these depth tests are various paper towels positioned to help identify the source of any potential leaks or faulty parts. Since this factor of safety is lower, we also perform additional visual inspections for cracking or yielding, and use Finite Element Analysis (FEA) to ensure a minimum FOS of 5.3.

Electrical unit testing involves bench-testing of all custom PCBs with an oscilloscope, multimeter, and other relevant electrical tools to ensure boards meet design specifications such as communication integrity requirements and sustained output under load with noise limitations. To measure total vehicle current draw, we used a current shunt and oscilloscope, which informed decisions around thruster current limiting and custom PCB design and validation. Software testing involves automated tests run by a linter to check for code style and syntax, as well as a code review process to ensure that everything works. The code review involves analyzing the code and testing all of the changes to ensure that they work as expected and do not interfere with other systems.

After a component or design change passes unit tests, we typically test it in its applicable subsystem on land to ensure basic mechanical, electrical, and software integration functions as intended. The subsystem integration testing was extremely valuable in troubleshooting manipulator issues, where we were able to identify and address issues such as i2c manipulator communication failures caused by a specific hardware-software interaction, frame-manipulator interface rigidity issues, and perform pneumatic system validation.

After basic unit testing and land integration testing, we integrate a component into our weekly full-system testing. Our weekly full system in-pool tests set objectives around operational safety and effectiveness, and fully integrated testing of changes and additions to benchmark against realistic simulated task missions. The tasks were primarily testing manipulator function, camera vision, and software programs. For example, for the crab identification task, images of the crabs were taken using the ROV camera under a variety of simulated environmental changes, for example varying the depth or using a higher light contrast, in order to ensure that the detection system would be robust. Prior to Dory's construction, prototype operational testing of smaller components such as the manipulators and camera enclosures, along with software programs such as the crab detection AI, were initially performed on CWRUbotix's previous ROV model, Fried Rice. This allowed for preliminary mechanical and software design validation and confirmed subsystem-integration before being transferred to Dory's model, which in turn provided rapid prototyping and faster testing.

These integrated tests provide extremely valuable information that directly informs system design and prototype evaluation. Moreover, the tests would be followed up during a weekly full team meeting, in which the test outcomes and objectives of the week to complete for the next test would be discussed. During the team meetings, all members were encouraged to attend and contribute ideas for design changes or other solutions.

Accounting and Budget

The CWRUbotix ROV team's budget is allocated each year out of the larger CWRUbotix organization's funds, which are split between each of the competition teams operating under the CWRUbotix umbrella. Specifically, the ROV team is allocated a portion of each relevant major funding source, as some funds are earmarked for specific purposes (e.g. robot parts or travel). Under this system, non-specialized purchases, such as general tools or lab safety equipment, come out of the overall CWRUbotix budget and not the ROV team's. These are therefore not included in this document. Projected budget requirements are estimated through an analysis of the previous year's spending in conjunction with any expected significant changes such as expensive materials or differences in travel costs. For a detailed view of the CWRUbotix ROV team's budget and project costing, see Appendix C.

Acknowledgments

Thank you to CWRUbotix's faculty advisor, Professor Richard Bachmann, at CWRU's Mechanical and Aerospace Engineering department for mentoring and support. Thank you also to the other professors and alumni who provided us with feedback during our design reviews. Thank you to the staff of the Veale Recreation Center, especially aquatics manager Brie Zola and Donnell Pool lifeguard Sofia Wilhelm, for allowing us to test our ROV in their pool. Thank you to our generous sponsors: the Case Alumni Association, Sears think[box], the Gene Haas foundation, Altium, and Dassault Systèmes for financial support and software licenses. Thank you to Teledyne Marine for their generous donation of two Blackfly S GigE machine vision cameras, and to Luxonis for their discount on OAK camera systems. Finally, thank you to MATE Inspiration for Innovation, MTS, and all the staff and volunteers of the World Championships for this incredible opportunity to present our work to the marine technology community.

References

[1] *AnyCubic Kobra 2 Max*. en. URL: `https://store.anycubic.com/products/kobra-2-max`.
[2] *Blackfly S GigE*. URL: `https://www.teledynevisionsolutions.com/products/blackfly-s-gige/`.
[3] *BlueRobotics T200 Thrusters*. en-US. URL: `https://bluerobotics.com/store/thrusters/t100-t200-thrusters/t200-thruster-r2-rp/`.
[4] *OAK-FFC 4P*. en. URL: `https://shop.luxonis.com/products/oak-ffc-4p`.

Appendices

Appendix A

Construction

- Ensure that the power is off and that the pneumatic line is depressurized
- Check that the inside of the e-bay is dry
- Make sure there are no loose wires and connections to penetrators are secure
- Check that the sealing surface is clean and entirely free of debris
- Lubricate O-ring as needed - use latex gloves to avoid skin contact
- Visually inspect the O-ring to confirm a good seal

After closing the e-bay:

- Ensure the e-bay lid is centered
- Ensure the latches of the e-bay are secured

Disassembly

- Ensure that the power is off and the pneumatic line is depressurized
- Dry the lid as best as possible, especially on the outer edges around the O-ring to prevent water from entering the e-bay

After opening the e-bay:

- Set the lid to the side so it is not resting on the penetrators
- Disconnect Ethernet between the lid penetrators and electronics to avoid electrical hazards
- Ensure that the e-bay is dry

Appendix B

Pre-Dive

1. Ensure that the power is off
2. Make sure the e-bay lid is sealed.
3. Verify poolside and ROV-side strain relief are secure
4. Connect control box to power supply and compressor
5. Unspool the tether and check for tangles
6. Verify electrical and pneumatic connections are secure
7. Check that the thrusters are free from obstructions
8. Turn on power to control station
9. Confirm that the surface computer has connected to the ROV
10. Verify all camera streams are displayed on the surface computer
11. Operator calls "Arming" and arms the ROV
12. Verify all thrusters and manipulators are functioning correctly
13. Disarm the ROV

Launch

1. Verify the ROV is disarmed
2. One person grab the ROV by the tether strain relief and put it in the water
3. Check that no bubbles are escaping from the e-bay. If there are bubbles, follow the Leak procedure.
4. If no bubbles are present, the poolside crew calls "Ready to arm"
5. Operator calls "Arming" and arms the ROV

Recovery

1. Operator calls "surfacing" as the pilot directs the ROV to the poolside surface
2. As the ROV reaches the surface, operator disarms the ROV and calls "Disarmed"
3. A poolside crew member pulls the ROV out of the pool by the tether strain relief and sets it down on the poolside
4. Operator turns power off from control station
5. Poolside crew checks for water in e-bay before relaunch or completion

Leak

1. Crew member calls "Leak"
2. Operator hits the emergency stop button in control station
3. Tether manager uses the tether to pull the ROV to the poolside
4. Visually check for water in e-bay. If water is present, remove the lid and dry all components with towels.
5. Check for corrosion on all electronics
6. Ensure all entry points are watertight with hydrophobic grease
7. After drying, test full system to ensure complete functionality

Appendix C

Budget Category	Sub-Category	Item Description	Type	Worth	Budget	Spent
Electrical	Onboard Hardware	Voltage Rugulators, Connectors	Purchased	$108.07	$600.00	$108.07
		T200 Thrusters x8, ESCs x8 RPi	Reused	$2,622.62	-	-
		POE Switch	Reused	$94.99	-	-
	Custom PCBs	Board manufacturing, Components	Purchased	$291.01	$3,300.00	$291.01
	Float Electronics	Motor, Batteries, Depth sensor	Reused	$192.94	$300.00	-
		Float Electronics Board	Reused	$239.57	-	$239.57
	Control Station Electronics	Router, Monitor, Control laptop	Reused	$1,459.71	$500.00	-
		Power Supply	Purchased	$453.00	-	$453.00
		Tether	Reused	$150.00	-	-
		Control station pneumatics, E-Stop button	Reused	$115.00	-	-
		USB dock	Reused	$70.00	-	-
	Spare COTS Components	Navigator Flight Controller	Purchased	$242.25	$300.00	$242.25
		T200 Thrusters x2	Reused	$516.00	-	-
Mechanical	Manufacturing	Waterjet machine time	Purchased	$18.60	$150.00	$18.60
	Fasteners and Mechanical Hardware	Nuts/Bolts, Misc mech components	Purchased	$510.10	$500.00	$510.10
		E-bay latches	Reused	$252.00	-	-
	Waterproofing	Epoxy	Purchased	$26.99	$200.00	$26.99
		Penetrators, Leftover o-rings	Reused	$171.88	-	-
	Raw Material and Stock	Sheet metal for frame and ebay, stock for enclosure endcaps, 3D Printer Filament	Purchased	$584.66	$550.00	$584.66
		Control station box	Reused	$169.95	-	-
		Onboard pneumatic pistons	Reused	$128.00	-	-
	Float Components	PCB Tubing	Purchased	$30.21	$100.00	$30.21
		Aluminum stock, Tube, O-Rings/X-Rings, Rods, Fasteners, Waterproof connector	Reused	$392.25	-	-

Sensors and Testing Hardware	Cameras, Sensors	Pressure Sensor	Purchased	$90.00	$1,000.00	$90.00
		Photosphere Components, FLIR Blackfly cameras x2	Reused	$1,549.98	-	-
		Luxonis stereo camera	Reused	$550.00	-	-
	Testing Hardware and Supplies	Pressure Test Chamber	Purchased	$219.06	$550.00	$219.06
		Sealants, epoxies, etc.	Purchased	$97.86	-	$97.86
Production Costs				**$11,248.84**	**$8,050.00**	**$2,911.38**
Competition Expenses	Registration and Fluid Power Quiz	Competition registration fees, Fluid power quiz	Purchased	$675.00	$675.00	$675.00
	Document Printing	Marketing display, Tech docs	Purchased	$150.00	$150.00	-
Miscellaneous	Prop Mockups	Specialized Fittings and Items for Props	Purchased	$26.05	$150.00	$26.05
	Tools and Equipment	3D Printer Replacement Nozzles, Tweezer, Sockets, etc.	Purchased	$293.39	$500.00	$293.39
		Underwater cam to document pool tests	Reused	$397.48	-	-
	Tax and Shipping	Tax and shipping on all purchases	Purchased	$700.97	$1,500.00	$700.97
	Pool Time	Lifeguard for weekly pool tests	Purchased	$728.00	$730.00	$728.00
Lodging and Travel (Internationals)	Housing	Lodging for the team in St John's, NL	Purchased	$2,200.00	$2,500.00	$2,200.00
	Gas	Transportation around St John's	Purchased	$300.00	$10,000.00	-
	Flights	Airfares for travel to and from St John's	Purchased	$8,390.09	-	$8,390.09
Administrative Costs				**$13,860.98**	**$16,205.00**	**$13,013.50**
Total Costs				**$25,109.82**	**$24,255.00**	**$15,924.88**